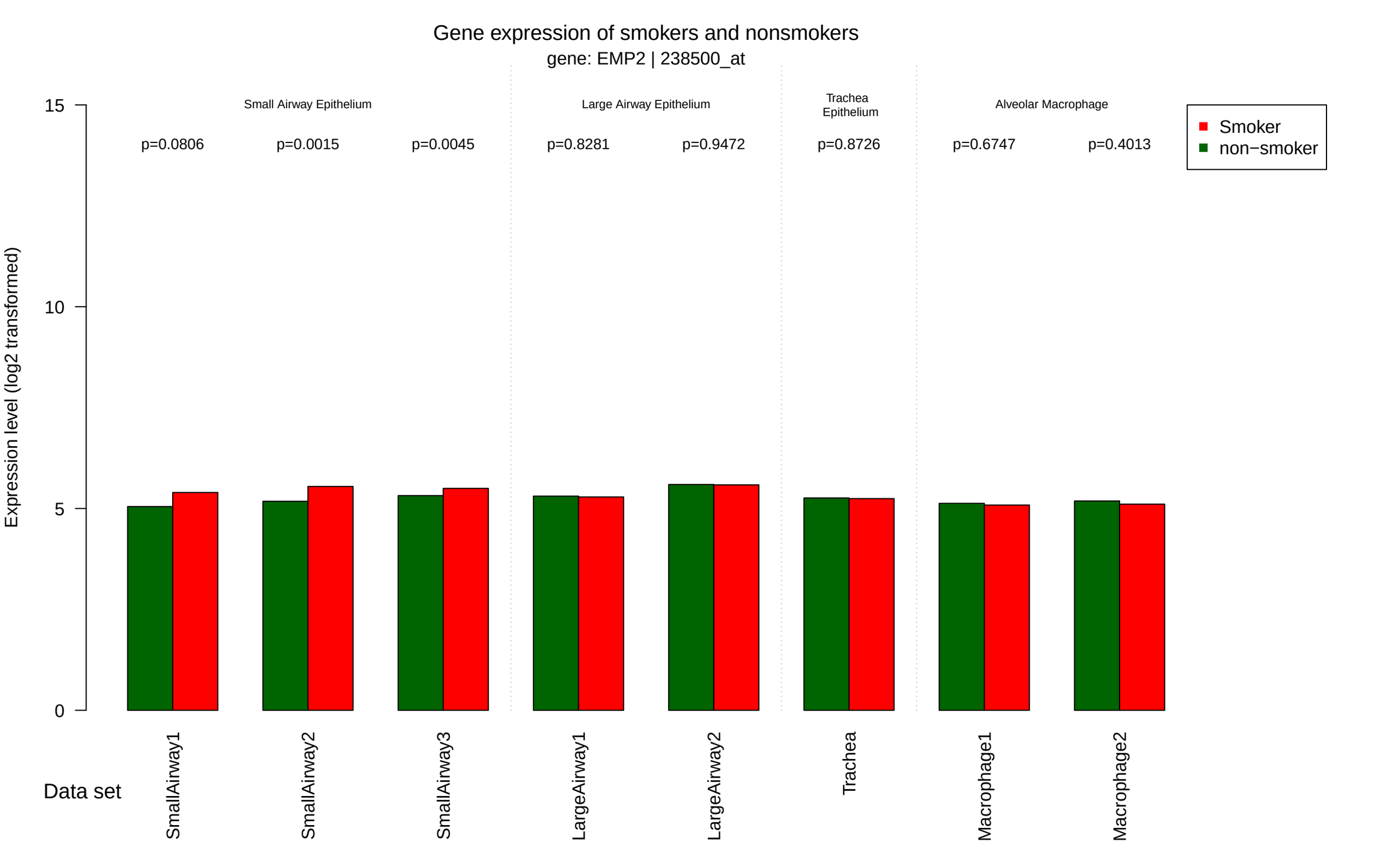
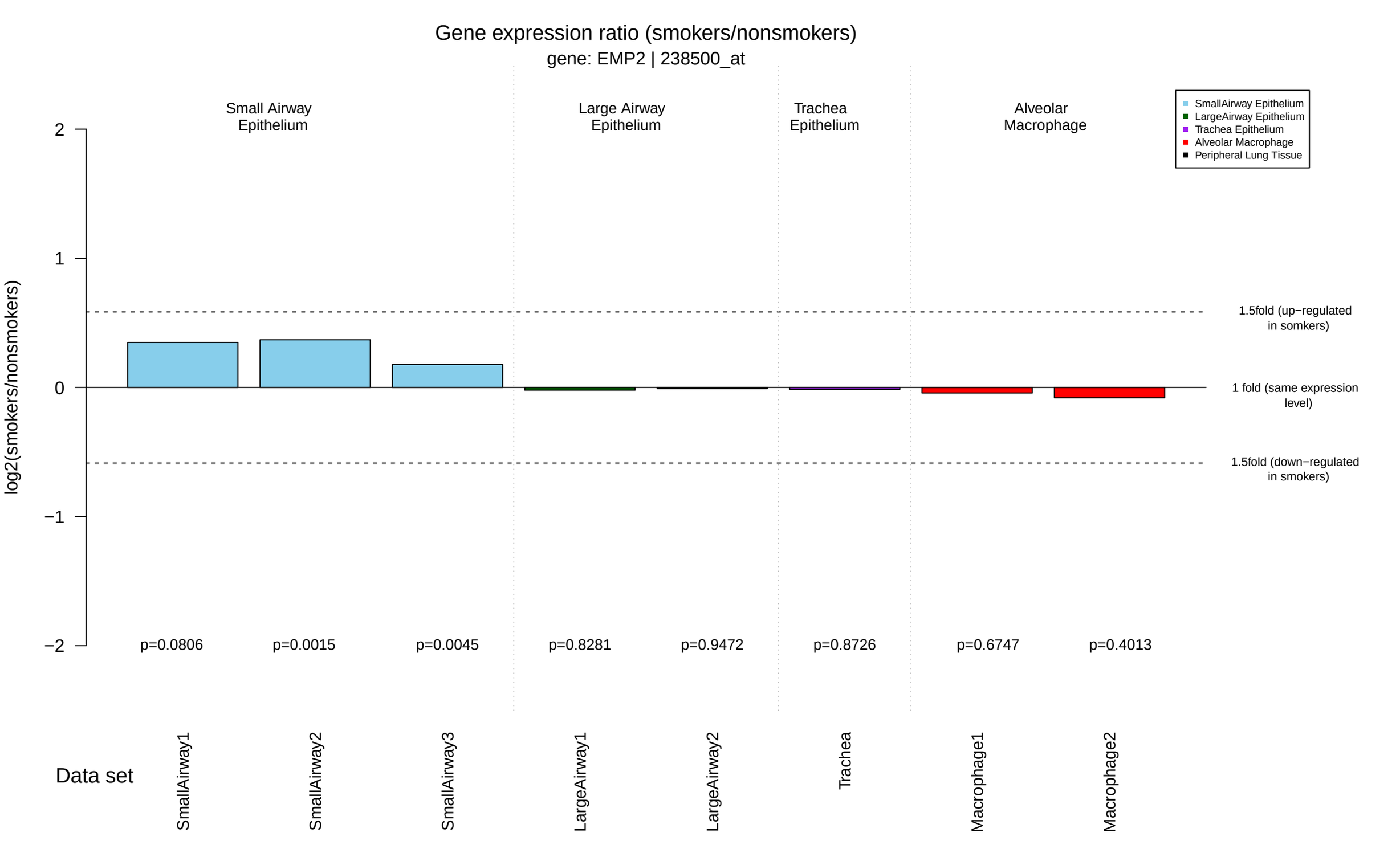
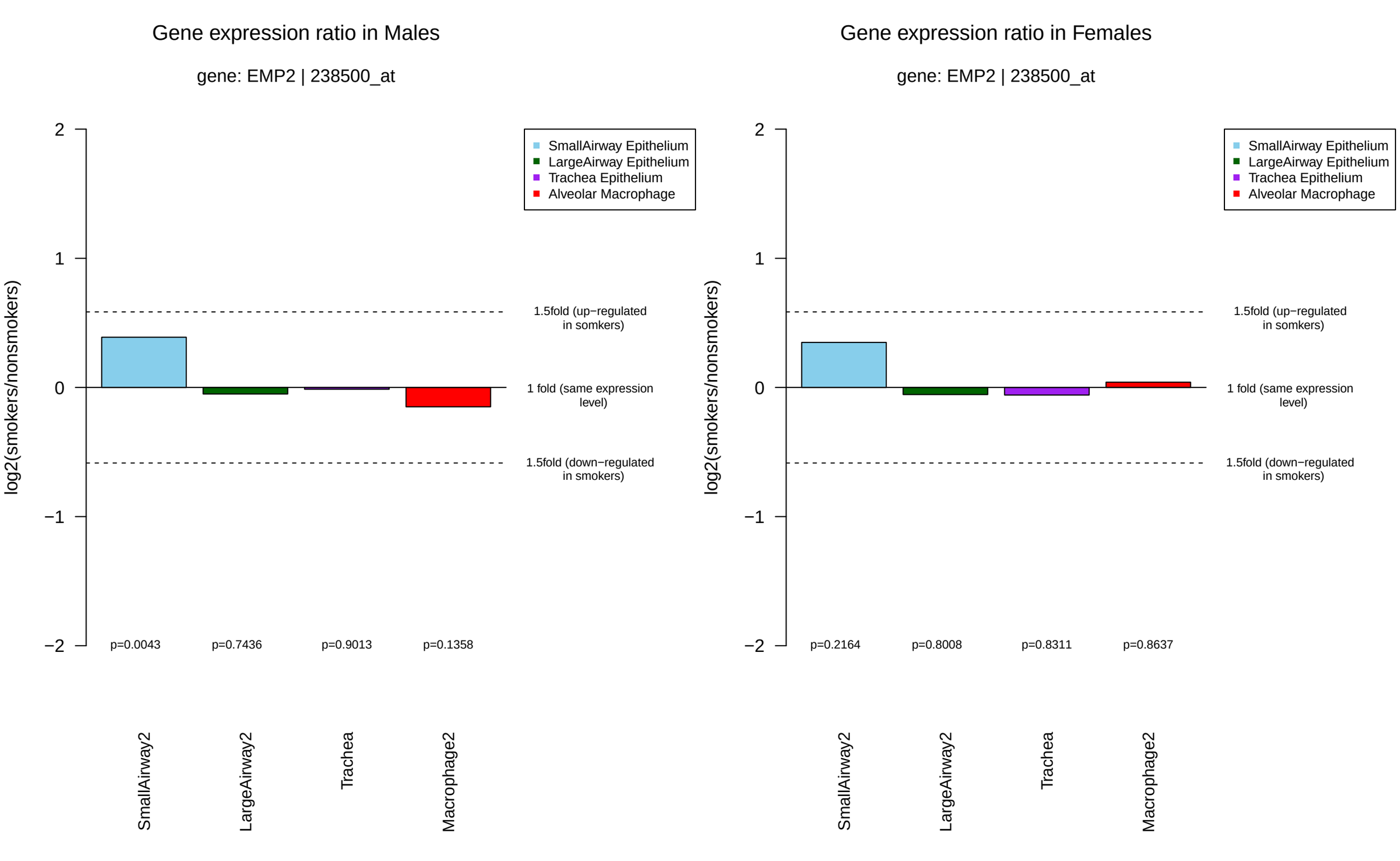
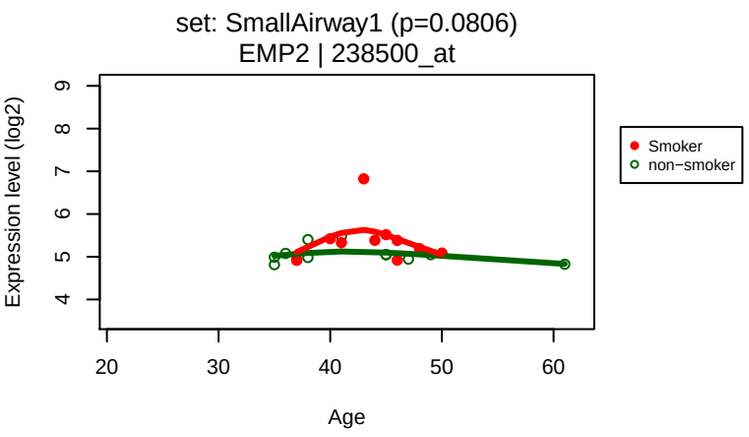
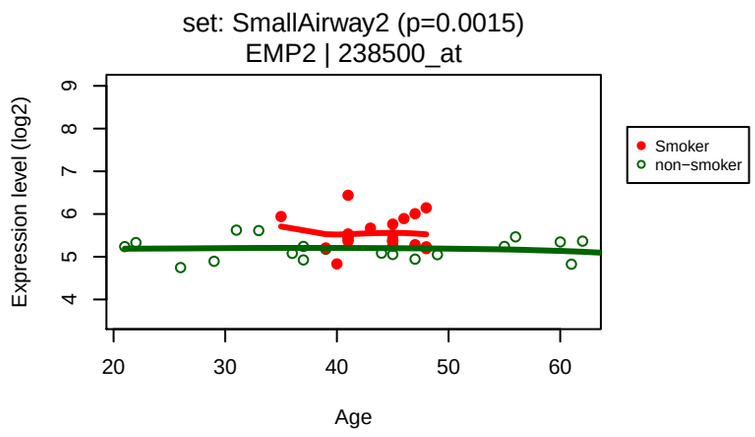
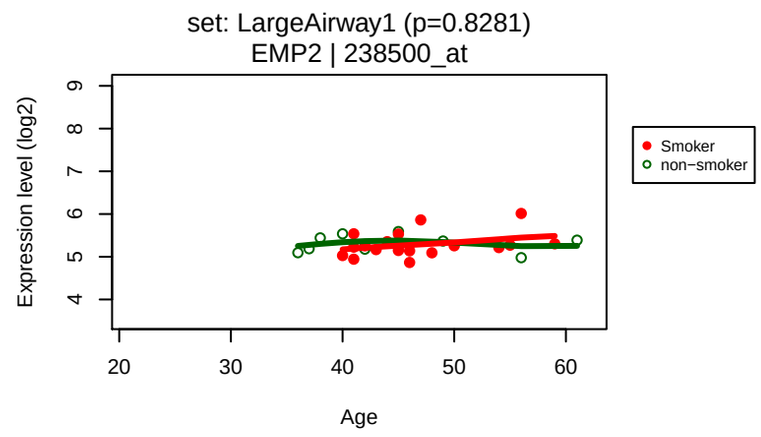

Gene expression of smokers and nonsmokers
gene: EMP2 | 238500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0806
p=0.0015
p=0.0045
p=0.8281
p=0.9472
p=0.8726
p=0.6747
p=0.4013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

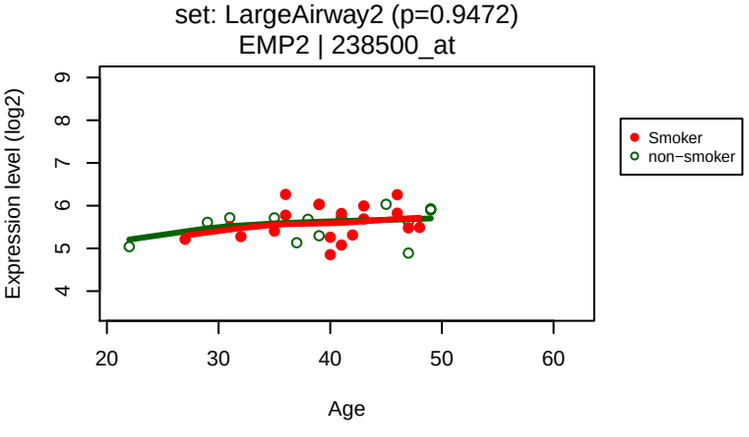

Gene expression ratio (smokers/nonsmokers)
gene: EMP2 | 238500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0806
p=0.0015
p=0.0045
p=0.8281
p=0.9472
p=0.8726
p=0.6747
p=0.4013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

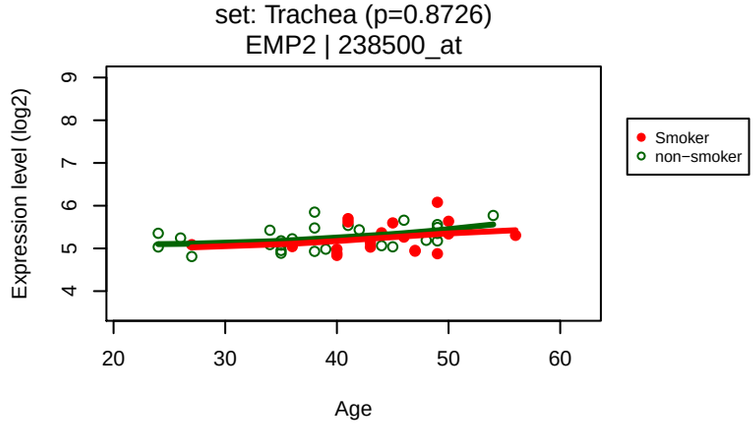

Gene expression ratio in Males
gene: EMP2 | 238500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0043
p=0.7436
p=0.9013
p=0.1358
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EMP2 | 238500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2164
p=0.8008
p=0.8311
p=0.8637
SmallAirway2
LargeAirway2
Trachea
Macrophage2

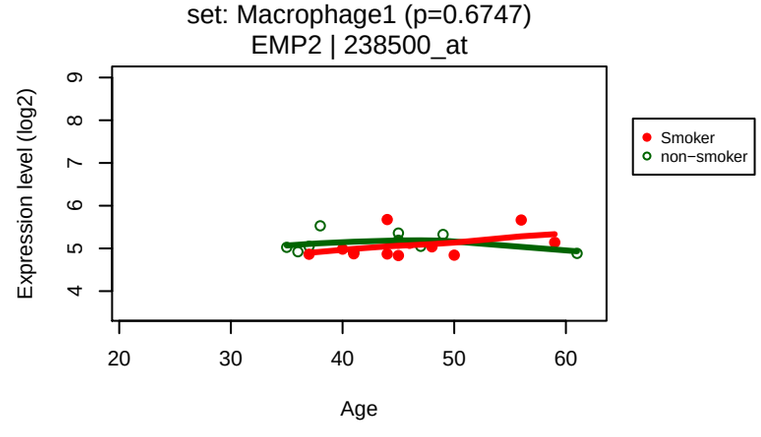

set: SmallAirway1 (p=0.0806)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker

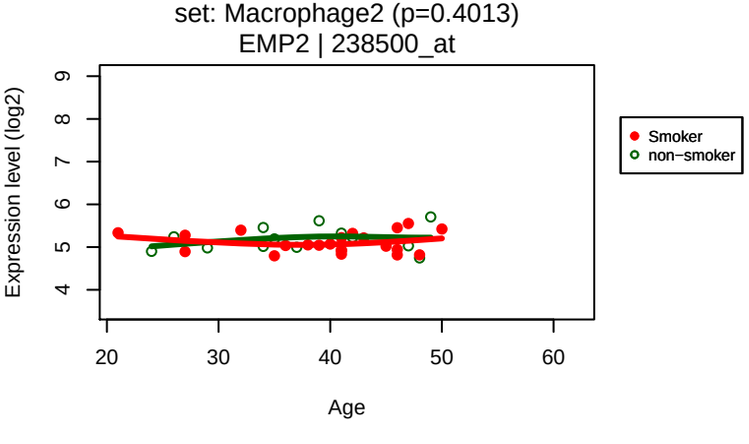

set: SmallAirway2 (p=0.0015)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker

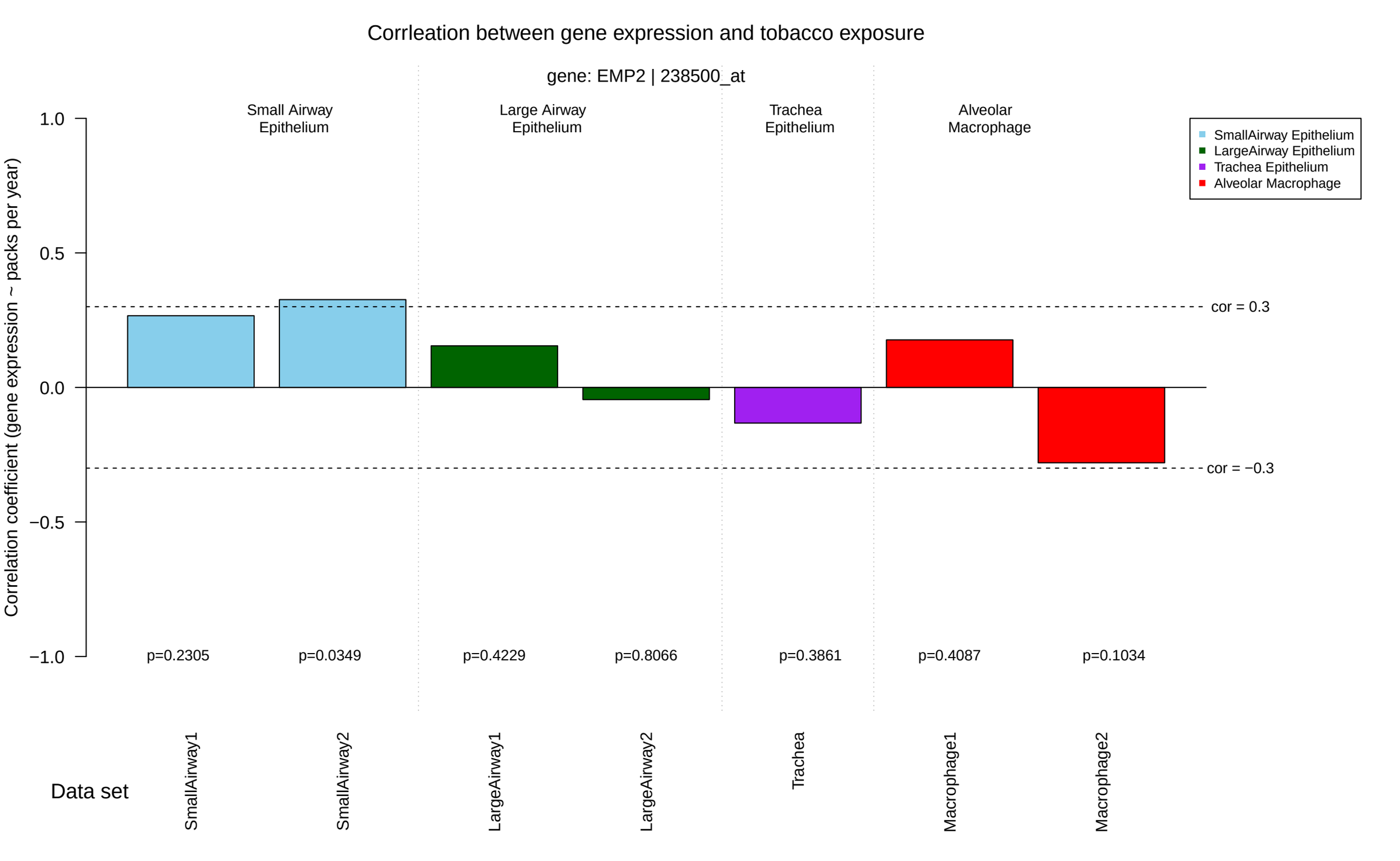

set: LargeAirway1 (p=0.8281)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9472)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8726)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6747)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4013)
EMP2 | 238500_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EMP2 | 238500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2305
p=0.0349
p=0.4229
p=0.8066
p=0.3861
p=0.4087
p=0.1034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2